

Marginal objective	$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(x)} \left[\ v_t(x) - u_t(x)\ ^2 \right]$ (Eq. 5)	$\mathcal{L}_{\text{RFM}}(\theta) = \mathbb{E}_{t,p_t(x)} \left[\ v_t(x) - u_t(x)\ _g^2 \right]$ (Eq. 3)
Conditional objective	$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1), p_t(x x_1)} \left[\ v_t(x) - u_t(x x_1)\ ^2 \right]$ (Eq. 9)	$\mathcal{L}_{\text{RCFM}}(\theta) = \mathbb{E}_{t,q(x_1), p_t(x x_1)} \left[\ v_t(x) - u_t(x x_1)\ _g^2 \right]$ (Eq. 8)
Conditional velocity u_t	$u_t(x x_1) = \frac{x_1 - (1 - \sigma_{\min})x}{1 - (1 - \sigma_{\min})t}$ (Eq. 21)	$u_t(x x_1) = \frac{d \log \kappa(t)}{dt} d(x, x_1) \times \frac{\nabla d(x, x_1)}{\ \nabla d(x, x_1)\ _g^2}$ (Eq. 13)
Conditional objective explicit form	$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1), p(x_0)} \left[\ v_t(\psi_t(x_0)) - (x_1 - (1 - \sigma_{\min})x_0)\ ^2 \right]$ (Eq. 23)	$\mathcal{L}_{\text{RCFM}}(\theta) = \mathbb{E}_{t,q(x_1), p(x_0)} \left[\ v_t(x_t) + d(x_0, x_1) \times \frac{\nabla d(x_t, x_1)}{\ \nabla d(x_t, x_1)\ _g^2} \ ^2 \right]$ (Eq. 14)
x_t definition	$x_t = \psi_t(x_0) = [1 - (1 - \sigma_{\min})t] x_0 + t x_1$ (Eq. 22)	$x_t = \exp_{x_1}(\kappa(t) \log_{x_1}(x_0)),$ $t \in [0, 1]$ (Eq. 15)